

CSE Core Subjects: Quick Revision Notes

Essential Concepts for Interviews and Exams

1. Data Structures & Algorithms (DSA)

DSA is the foundation of efficient problem solving.

- **Arrays:** Linear data structure stored in contiguous memory.
- **Linked List:** Elements (nodes) are connected using pointers.
- **Stack (LIFO):** Last In First Out (e.g., undo function).
- **Queue (FIFO):** First In First Out (e.g., printer tasks).

Sorting Algorithms Time Complexity:

Algorithm	Best Case	Worst Case
Bubble Sort	$O(n)$	$O(n^2)$
Merge Sort	$O(n \log n)$	$O(n \log n)$
Quick Sort	$O(n \log n)$	$O(n^2)$

2. Database Management System (DBMS)

DBMS is software for storing and retrieving user data while considering safety measures.

ACID Properties:

- **Atomicity:** Entire transaction happens at once or not at all.
- **Consistency:** Database must be consistent before and after transaction.

- **Isolation:** Multiple transactions occur independently.
- **Durability:** Changes are saved permanently in the database.

3. Object-Oriented Programming (OOP)

Focuses on objects rather than logic/functions.

- **Inheritance:** Child class acquiring properties of parent class.
- **Polymorphism:** One interface, multiple forms (Overloading vs Overriding).
- **Encapsulation:** Wrapping data and code into a single unit (Class).
- **Abstraction:** Hiding internal details and showing only functionality.

4. Operating Systems (OS)

Key Concepts:

- **Process:** A program in execution.
- **Thread:** A lightweight process.
- **Deadlock:** A situation where processes are stuck waiting for each other.
- **Virtual Memory:** Allows execution of processes that may not be completely in memory.

5. Computer Networks

OSI Model Layers:

1. Physical Layer
2. Data Link Layer
3. Network Layer
4. Transport Layer
5. Session Layer
6. Presentation Layer
7. Application Layer